

Angad Singh Kochhar

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EDUCATION

Purdue University

West Lafayette, IN, USA

Master of Science in Engineering, Robotics / Interdisciplinary Engineering

Expected Dec. 2026

GPA: **3.84/4.00**

Selected Graduate Coursework: Reinforcement Learning, Multi-Agent Autonomy & Control, Artificial Intelligence, Robotics Kinematics & Dynamics, Linear Algebra, Mechatronics, Embedded Systems, Industrial IoT for Smart Manufacturing

Guru Gobind Singh Indraprastha University

New Delhi, India

Bachelor of Technology, Mechanical & Automation Engineering

Sep. 2020

84.3%

RESEARCH INTERESTS

Reinforcement Learning, Embodied Intelligence, Multi-Agent Systems, Robot Learning, Emergent Behavior, Autonomous Systems

RESEARCH EXPERIENCE

SCALE Robotics Lab, Purdue University

May 2026 – Present

Student Researcher — Recreating Emergent Tool Use in Multi-Agent Reinforcement Learning

West Lafayette, IN

Advisor: Prof. Rohan R. Paleja

- Investigating emergent strategies in multi-agent reinforcement learning by developing physics-based environments in MuJoCo inspired by prior work on emergent tool use.
- Developing a configurable environment containing agents, physical objects, and interactive tools to study how simple reward structures and reinforcement-learning policies can produce complex behaviors.
- Building foundations for experiments using Proximal Policy Optimization (PPO), with planned evaluation of agent strategies and emergent interactions in increasingly complex environments.

SCALE Robotics Lab, Purdue University

May 2026 – July 2026

Student Researcher — Autonomous Connect-4 with Boston Dynamics Spot

West Lafayette, IN

Faculty Advisor: Prof. Rohan R. Paleja — Project Lead: Daniel Chen

- Developed and tested an autonomous Connect-4 system for Boston Dynamics Spot integrating computer vision, manipulation, game-state estimation, and decision-making for human-robot gameplay.
- Integrated YOLO-based board perception and AprilTag localization with a minimax game-playing algorithm to convert visual board states into autonomous manipulation actions.
- Designed failure-detection and recovery mechanisms for chip acquisition and manipulation, including gripper-state monitoring, visual confirmation of human and robot moves, and automatic retry behavior.
- Improved system reliability through iterative testing, achieving greater than 90% chip-detection success and greater than 85% successful chip placement across more than 100 trials.

Purdue University

Aug. 2025 – Nov. 2025

Research Student — Spacecraft Autonomy: Multi-Agent Coordination

West Lafayette, IN

ENGR 59600-008 — Autonomy, IoT, & Robotics Project — Faculty Supervisor: Prof. Kenshiro Oguri

- Developed a MATLAB simulation framework for spacecraft formation flying in low Earth orbit using the Clohessy-Wiltshire linearized relative-motion equations.
- Extended the simulation to incorporate J2 perturbations and investigated their effects on multi-spacecraft formation behavior.
- Explored PID-based formation maintenance and developed simulation and analysis tools for studying autonomous coordination in multi-spacecraft systems.

Impact of Stubborn Truth Agents on Convergence in Opinion Dynamics under Random Edge Failures Spring 2026

AAE 59000 — Multi-Agent Autonomy & Control, Purdue University

- Investigated truth propagation in distributed multi-agent networks using opinion-dynamics models with stubborn agents, random communication failures, and varying network connectivity.
- Implemented simulation experiments in Python/MATLAB to evaluate convergence behavior, showing that stubborn agents drive networks toward a fixed truth value while additional stubborn agents accelerate convergence.
- Conducted a semester-long literature study of distributed consensus and multi-agent systems and presented the work in a research-paper format.

ModifiedNAFNet: Lightweight Image Restoration

Spring 2025

ECE 57000 — Artificial Intelligence, Purdue University

- Reimplemented and modified the NAFNet image-restoration architecture to create a lightweight variant suitable for limited-resource environments such as Google Colab.
- Reduced model width, stages, and block complexity while retaining useful denoising performance, achieving **33.66 dB PSNR** and **0.9400 SSIM** on the SIDD dataset with approximately two hours of training.
- Studied the original research paper and implementation, developed the modified architecture in PyTorch, and presented the project through an ICML-style paper, peer-review exercise, and class presentation.

Low-Cost Indoor Positioning Using Combined Inertial Navigation and Wireless Positioning Spring 2026

ECE 56800 — Embedded Systems, Purdue University

- Developed the UWB subsystem for a sub-\$100 indoor positioning prototype combining Ultra-Wideband ranging with inertial sensing.
- Investigated multipath and wall-reflection effects through experimental testing, including validation in an anechoic chamber, and implemented spike-thresholding and filtering methods in C++.
- Redesigned UWB sensor placement and antenna orientation to improve ranging stability and reduce reflection-induced errors.

DED Machine Monitoring Using Acoustic Machine Learning

Spring 2026

ME 59700 — Industrial IoT for Smart Manufacturing, Purdue University

- Led a three-person team investigating abnormal acoustic signatures from a FormAlloy L2 directed-energy-deposition system in Purdue's Materials and Manufacturing Research Laboratory.
- Developed a Random Forest model using acoustic sensor data to identify machine-component operating states and support anomaly localization.
- Designed isolated machine tests to obtain cleaner component-level acoustic signatures; analysis traced a major anomalous signal to the machine's moving stage and motivated subsequent mechanical inspection.

Gladiator E-Cove: Battle for Logistics

Spring 2025

ME 58800 — Mechatronics, Purdue University

- Designed and fabricated a dual-flywheel projectile launcher using projectile-motion and flywheel-energy calculations, achieving approximately **85% delivery accuracy across 100+ shots**.
- Built and refined an IR sensing system and implemented finite-state-machine logic and ESP32 communication for an autonomous two-robot logistics system.
- Developed a motor-startup and RPM-control strategy to overcome low-speed stall behavior; the team earned the **highest score in the final competition**.

PROFESSIONAL EXPERIENCE

Static Systems Electronics Pvt. Ltd.

Sep. 2020 – Dec. 2024

Assistant Mechanical Engineer

Gurgaon, India

- Supported the design, fabrication, fit, and installation of customized physical-security equipment for government and private-sector clients, ensuring products met customer requirements and applicable security specifications.
- Served as a field engineer on customer sites, guiding installation teams, troubleshooting mechanical and installation issues, and supporting commissioning of security systems.
- Coordinated directly with customers on equipment operation, installation requirements, and safe use of security systems.
- Supported deployment of electric-fencing systems using NEMTEK equipment, applying manufacturer training to oversee field installations and provide technical guidance.

TECHNICAL SKILLS

Programming: Python, C/C++, MATLAB, Lua

Robotics & Simulation: MuJoCo, ROS 2, CoppeliaSim, RViz, Boston Dynamics Spot, UR5e

Machine Learning: PyTorch, torchvision, scikit-learn, YOLO, Random Forest

Embedded Systems & Hardware: ESP32, Raspberry Pi, UWB, IMU, Arduino, sensor integration

Tools: Git/GitHub, Linux/Ubuntu, VS Code, LaTeX/Overleaf, Arduino IDE, KiCad

LEADERSHIP & ACTIVITIES

Amity Greenhorns Military Camp

Summer 2019

Platoon Commander

India

- Selected to lead a platoon during mandatory undergraduate military training; coordinated the group during camp activities and final parade preparation.
- Led the platoon to first place in the final-day parade competition.